PRODUCT REQUIREMENTS DOCUMENT

JIGAWA STATE ELECTION SITUATION ROOM (JESR)

Integrated Real-Time Election Monitoring & Reporting Platform

v2.0 - Extended to Polling Unit (PU) Level Reporting

Document Type	Product Requirements Document (PRD)
Product Name	Jigawa State Election Situation Room (JESR)
Version	2.0 - Polling Unit Level Reporting Update
Change from v1.0	Reporting granularity extended from Ward to Polling Unit (PU). Ward remains the supervisory rollup level.
Hierarchy	State > Senatorial District > LGA > Ward > Polling Unit (PU)
Coverage	Jigawa State - 27 LGAs, 287 Wards, ~3,459 Polling Units (approx. 12 PUs per Ward avg.)
Platform	Web (Laravel/Livewire), Android & iOS (Flutter), SMS Gateway
Status	Active Requirements - v2.0 Update

Confidential - For Official Use Only

1. Executive Summary

The Jigawa State Election Situation Room (JESR) is a purpose-built, multi-platform digital system for collecting, monitoring, analysing, and responding to election data across Jigawa State. Version 2.0 of this PRD extends the reporting granularity from Ward level to Polling Unit (PU) level — the atomic unit of Nigeria's electoral geography as defined by INEC.

> **PU-Level:** Under INEC's structure, each Ward is sub-divided into Polling Units (PUs). A PU is where voters physically cast their ballot. Jigawa State has approximately 3,459 registered PUs across 287 Wards in 27 LGAs. Reporting at PU level gives the Situation Room the most granular, verifiable, and actionable picture of election-day activity.

JESR operates across four channels: Web Application, Android App, iOS App, and SMS Gateway. Each Polling Unit Agent (PUA) is assigned to a specific, single PU. Ward-level and LGA-level data are automatically aggregated from PU submissions. Supervisors and Super Admins continue to view data at Ward, LGA, Senatorial District, and State levels — now enriched by PU-resolution drill-down.

Key Outcomes Addressed in v2.0

- PU-level structured data capture: each submission is uniquely tied to a Polling Unit code (INEC PU Code format: AA/BB/CCC/DDD)
- Automatic aggregation: PU data rolls up to Ward, Ward to LGA, LGA to Senatorial District, and to State level
- PU agent assignment: each Ward may have multiple Polling Unit Agents, one per PU
- PU-resolution dashboards: all dashboard panels support PU-level filter and drill-down
- Updated SMS format: PU code replaces Ward code as the primary location identifier
- PU coverage tracking: real-time visibility of which PUs have reported and which remain outstanding

2. Electoral Hierarchy & Reporting Structure

2.1 Nigeria INEC Reporting Hierarchy

INEC structures election geography in a strict five-tier hierarchy. JESR v2.0 captures data at every level of this hierarchy, with Polling Unit as the atomic input point:

Tier	Level	Jigawa Count	JESR Role
1	State	1	Super Admin dashboard: full state overview and final authority
2	Senatorial District	3 (North-West, North-East, North-Central zones)	Rollup analytics panel; used for candidate district aggregates
3	Local Government Area (LGA)	27 LGAs	Supervisor assignment scope; LGA-level choropleth dashboard
4	Ward	287 Electoral Wards	Supervisory rollup; Ward-level report aggregate and verification
5 (NEW)	Polling Unit (PU)	~3,459 PUs (avg. ~12 per Ward)	ATOMIC REPORTING LEVEL: one submission per PU per Polling Unit Agent (PUA)

2.2 Polling Unit Agent (PUA) Assignment Model

> **PU-Level:** In v2.0, the role "Ward Agent" is renamed to "Polling Unit Agent (PUA)". Each PUA is assigned to exactly one Polling Unit. A Ward with 10 PUs has 10 PUAs. Supervisors oversee all PUs within their assigned LGA(s).

Assignment rules:

- One PUA per Polling Unit (1:1 assignment). A PUA cannot submit for a PU they are not assigned to.
- A Ward may have between 1 and 40+ PUs. All PUs within a Ward are supervised by the same LGA Supervisor(s).
- PU assignment is set during agent enrolment by Super Admin and cannot be self-modified by the PUA.
- Super Admin may reassign a PUA to a different PU before election day (with full audit trail).
- If a PU has no assigned PUA on election day, the system flags it as "Unassigned PU" in the coverage dashboard.

2.3 Data Aggregation Flow

Flow	PU Report (PUA submits) --> Ward Aggregate (auto-sum of all PUs in Ward) --> LGA Aggregate --> Senatorial District Aggregate --> State Dashboard

Aggregation behaviour:

- Numeric fields (votes cast, valid ballots, invalid ballots, voter papers) are auto-summed from all submitted PU reports within a Ward.
- Status fields (security present, escalations) are surfaced at Ward level when any PU within the Ward has that flag set.
- Incomplete coverage: Ward aggregates are displayed with a "X of Y PUs reported" indicator to signal partial data.
- Only verified PU reports contribute to official Ward and LGA aggregates. Flagged and escalated reports are shown separately.

3. Problem Statement & Goals

3.1 Problem Statement

In addition to the operational gaps identified in v1.0, Ward-only reporting introduces a specific structural problem:

- Loss of granularity: Reporting at Ward level merges data from multiple Polling Units into a single submission, making it impossible to identify which specific PU had irregularities, security incidents, or vote count anomalies.
- Accountability gaps: When a Ward Agent submits one report for the whole Ward, it is unclear whether they visited all PUs or are estimating totals. PU-level reporting creates a verifiable, PU-by-PU audit trail.
- Incident localisation failure: Security incidents or ballot irregularities at a specific PU cannot be mapped or escalated at the precise PU location under Ward-only reporting.
- Coverage blind spots: Without PU-level tracking, it is impossible to know how many of a Ward's PUs have completed their count and how many remain outstanding.

3.2 Product Goals (v2.0 Updates)

#	Goal	Success Criteria
G1	PU-level real-time situational awareness	100% of submitted PU reports visible within 60 seconds; drill-down from State to PU in <= 3 clicks
G2	PU-specific structured data capture	Every report uniquely tied to a verified INEC PU Code; no report accepted without valid PU assignment
G3	Full PU coverage tracking	Dashboard shows real-time "X of Y PUs reported" per Ward and per LGA at all times
G4	Automatic Ward/LGA aggregation	All PU submissions auto-aggregate to Ward and LGA totals without manual intervention
G5	Offline-first for PU field conditions	PUA can draft and auto-sync PU reports with zero data loss on reconnect
G6	PU-resolution incident escalation	Escalation records include exact PU Code and location coordinates (if GPS available)
G7	SMS reporting at PU level	SMS format includes PU Code; parsing success rate >= 90% for correctly formatted messages
G8	PU-resolution print reporting	All 7 standard report types support PU-level filter and export

4. User Personas & Access Model

4.1 Layer 1 - Super Admin (State Situation Room)

Role	State-level monitoring authority with full system control across all LGAs, Wards, and Polling Units

Responsibilities (v2.0 additions):

- Manage PU master data: create, update, and deactivate Polling Unit records linked to Wards and LGAs
- Assign and reassign PUAs to specific Polling Units during the enrolment phase
- Monitor PU coverage dashboard: identify unassigned or non-reporting PUs before and during election day
- Generate and export PU-level reports across all seven standard dimensions
- All v1.0 Super Admin responsibilities remain unchanged

4.2 Layer 2 - Supervisor (Call Desk / LGA Coordinator)

Role	LGA-assigned monitoring authority overseeing all Polling Units within assigned LGA(s)

Responsibilities (v2.0 additions):

- Monitor PU-level Inbox: incoming reports are now at PU granularity, not Ward level
- Track PU coverage within assigned LGAs: see which PUs have submitted, which are late, which are flagged
- Escalate incidents at PU level: escalation records capture the specific PU Code and location
- All v1.0 Supervisor responsibilities remain unchanged

4.3 Layer 3 - Polling Unit Agent / PUA (Field Reporter)

Role	Election observer assigned to exactly one Polling Unit; responsible for submitting one PU-level report

Key changes from v1.0 Ward Agent model:

> **PU-Level:** The "Ward Agent" role is replaced by "Polling Unit Agent (PUA)". A PUA is pre-assigned to exactly one Polling Unit. Their form is pre-filled with LGA, Ward, AND Polling Unit. They submit one report per election for their PU. The correction window applies per PU report.

- PUA can only submit a report for their assigned Polling Unit — the system rejects cross-PU submissions
- PUA sees their own PU report and system messages only (not other PUs in the same Ward)
- PUA receives a confirmation with their PU Code and Report ID on submission

4.4 Role & Permission Matrix (v2.0)

Capability	Super Admin	Supervisor	Polling Unit Agent (PUA)
Create / Manage Accounts	Full	None	None
Manage PU Master Data	Full (create/update/deactivate PUs)	None	None
Assign PUA to PU	Full	None	None
Submit PU Report	Full (any PU)	If assigned to PU	Assigned PU only
View Reports	All PUs / Wards / LGAs	Scoped LGA (all PUs within)	Own PU report only
View PU Coverage Dashboard	Full (all LGAs)	Scoped (assigned LGA)	Own PU status only
Escalate Incidents	Full	Full (with PU Code)	None
Run & Print Reports (PU-level)	Full	Scoped	None
Send SMS	Full (to all PUs / groups)	Scoped by LGA	Receive only
View Audit Logs	All logs	Scoped logs	Own PU entry only
MFA Required	Mandatory	Recommended	Optional

5. Functional Requirements

5.1 Authentication & Access Control

- FR-AUTH-01: Three-role authentication: Super Admin (Layer 1), Supervisor (Layer 2), Polling Unit Agent / PUA (Layer 3).
- FR-AUTH-02: Super Admin accounts enforce MFA (SMS OTP or authenticator app).
- FR-AUTH-03: PUAs authenticate via phone number + password with device binding (phone + device ID).
- FR-AUTH-04: JWT tokens for session management with short expiry and refresh token rotation.
- FR-AUTH-05: PUA login is bound to their assigned Polling Unit. The system rejects any attempt to submit a report for a PU other than the one in their profile.
- FR-AUTH-06: Row-level data visibility enforced at the database query level per role, LGA, Ward, and Polling Unit assignment.

5.2 Polling Unit Master Data Management

> **PU-Level:** This is a new functional area introduced in v2.0. The PU master list must be loaded and maintained as a prerequisite to agent enrolment.

- FR-PU-01: Super Admin shall be able to create, update, and soft-delete Polling Unit records. Hard delete is not permitted — deletion is flagged for audit.
- FR-PU-02: Each PU record shall contain: PU Code (INEC format: SS/LGA/WardNum/PUNum e.g. JG/06/01/003), PU Name, Ward ID, LGA ID, Registered Voter Count, and optional GPS coordinates (latitude/longitude).
- FR-PU-03: PU codes shall be validated against the INEC PU code format pattern on entry.
- FR-PU-04: Super Admin can bulk-upload PU records via CSV. The system validates format and flags duplicates before import.
- FR-PU-05: Each PU displays its assignment status: Assigned (with agent name) or Unassigned on the PU management screen.
- FR-PU-06: Super Admin can assign or reassign a PUA to a PU. Reassignment requires a reason and is logged in the audit trail.

5.3 Data Capture - PU Report Form

Part A: Identity & Metadata (Auto-Populated at PU Level)

- Agent Name: pre-filled from authenticated account profile
- Local Government Area (LGA): pre-filled from account profile
- Ward: pre-filled from account profile

> **PU-Level:** Polling Unit (PU): pre-filled from the PUA's assigned PU record. Displays PU Code, PU Name, and Registered Voter Count. This field is READ-ONLY for PUAs — they cannot select a different PU.

- INEC PU Code: displayed prominently as read-only; included on submission confirmation
- Registered Voter Count: pre-filled from PU master data; shown as reference (non-editable)
- Time of Reporting: auto-generated server timestamp; manual override within correction window only, requiring audit log entry

Part B: Agents & Security

- Party Registered Agent: searchable dropdown from registered party agents for this PU (nullable)
- INEC Presiding Officer Name: free-text field (optional) — replaces INEC Agent Name to reflect PU-level terminology
- Number of Counted Voter Papers (Accredited Voters): integer >= 0
- Security Personnel Present: dropdown Yes / No (required)
- Security Type(s): multi-select when security_present = Yes | Police, Civil Defence, NSCDC, Army, Vigilante, Other

Part C: Voting Counts

- Time Voting Closed at this PU: time picker (nullable)
- Number of Persons Who Cast Votes: integer >= 0
- Voting Papers Left (Not Cast / Unused): integer >= 0
- Valid Ballot Papers: integer >= 0
- Invalid / Spoilt Ballot Papers: integer >= 0

> **PU-Level:** Turnout Rate (auto-calculated): (Number Cast / Registered Voter Count) x 100, displayed in real-time as the agent fills the form. Read-only; helps field agents spot entry errors before submission.

Part D: Candidate / Party Results (PU Level)

> **PU-Level:** This sub-section is NEW in v2.0. Party/candidate vote totals are captured at PU level, enabling granular result tracking and cross-verification with INEC form EC8A.

- Party Results Entry: for each registered party, the agent enters the number of votes scored. The form dynamically generates one row per active party from the master party list.
- Validation: Sum of all party votes must equal valid_ballot_papers. A real-time running total is shown. Discrepancy triggers a warning and soft-blocks submission until resolved or annotated.
- INEC Form Reference: optional field for the PUA to enter the EC8A form serial number for cross-referencing.

Part E: Media, Remarks & Submission

- Photo Attachments: 0-5 photos (JPEG, PNG). Recommended: photo of EC8A result sheet.
- Video Attachment: optional; 1 video max; maximum 50 MB
- Audio Note: optional; 1 clip max
- Remarks: free-text, maximum 1000 characters
- Save as Draft / Submit: correction window applies post-submission (default: 10 minutes, configurable)

5.3.1 Form Validation Rules (v2.0 Updates)

- FR-FORM-01: agent_name, lga_id, ward_id, polling_unit_id, and submission_timestamp are required. Form shall not submit without all five.
- FR-FORM-02: The polling_unit_id in the submission must match the PUA's assigned polling_unit_id in the users table. Mismatch is rejected by the API.

- FR-FORM-03: All numeric count fields accept only non-negative integers.
- FR-FORM-04: Cross-field check: (valid + invalid) <= number_cast_votes. Violation triggers warning and flags report for supervisor review.
- FR-FORM-05: Party result sum check: sum(party_votes) must equal valid_ballot_papers. Discrepancy shows running delta and soft-blocks until acknowledged.
- FR-FORM-06: If security_present = Yes, at least one security type must be selected.
- FR-FORM-07: number_cast_votes should be <= registered_voter_count (warning only; turnout > 100% is flagged automatically for escalation consideration).

5.4 Offline-First & Synchronisation

- FR-OFFLINE-01: Mobile app allows PUA to draft and save their PU report locally with no internet, using on-device SQLite.
- FR-OFFLINE-02: Offline drafts carry local timestamp (submitted_at_device), device metadata, and the pre-filled PU Code.
- FR-OFFLINE-03: On reconnection, app auto-submits pending draft via POST /api/reports (or batch endpoint).
- FR-OFFLINE-04: Server assigns authoritative timestamp but preserves device timestamp for audit.
- FR-OFFLINE-05: Conflicting submissions for the same PU ID are flagged for Supervisor review.
- FR-OFFLINE-06: Sync status and last sync time displayed prominently in the PUA app.
- FR-OFFLINE-07: Party result rows and media attachments are stored locally and synced with the report on reconnect.

5.5 SMS Gateway (v2.0 - PU Code Required)

Updated Inbound SMS Format

PU-Level: The SMS format is updated in v2.0 to include the full INEC PU Code as the primary location identifier. The Ward Code is no longer used in SMS submissions.

v2.0 SMS Format	JIG PU <PUCode> <AgentInitials> <PartyCode> <Accredited> <Valid> <Invalid> <Sec:Y/N> <SecCodes>

Example	JIG PU JG/06/01/003 MA APC 120 118 2 Y POLICE,VIG

- FR-SMS-01: Only PUAs with registered phone numbers are accepted as authenticated SMS senders.
- FR-SMS-02: The system validates the PU Code in the SMS against the sender's assigned PU. If they do not match, the SMS is rejected and an error reply is sent.
- FR-SMS-03: Successful parsing creates a pu_reports record (submitted_via = "sms") and sends confirmation: CONF PU <PUCode> <ReportID> Received. Thank you.
- FR-SMS-04: Parse failures trigger an auto-reply with the correct format and the sender's assigned PU Code.
- FR-SMS-05: Party-level results cannot be captured via SMS. SMS submissions create a partial report flagged for media/results follow-up.

Outbound SMS

- FR-SMS-06: Super Admin and Supervisors can send broadcasts to PUAs grouped by: all PUAs, PUAs by LGA, PUAs by Ward, or individual PU.
- FR-SMS-07: SMS delivery receipts stored and displayed in SMS management interface.
- FR-SMS-08: Failed SMS auto-retried up to 3 times with exponential backoff.

5.6 Supervisor Workflow (v2.0 Updates)

- FR-SUP-01: Supervisor Inbox shows PU-level reports sorted by severity (escalated > flagged > new). Each entry displays: PU Code, PU Name, Ward, Agent name, Timestamp, Submission method.
- FR-SUP-02: Quick actions per PU report: Verify, Flag (reason required), Escalate, Request Clarification, Call Agent.
- FR-SUP-03: Escalation records now include: PU Code, PU Name, and GPS coordinates if available in PU master data.
- FR-SUP-04: Supervisors see a "Ward Coverage Status" widget showing for each Ward in their LGA: total PUs, PUs submitted, PUs pending, PUs flagged/escalated.
- FR-SUP-05: "Missing PUs" alert: if a PU has not submitted by a configurable threshold time (default: 30 minutes before the correction window closes), the system auto-alerts the LGA Supervisor.

5.7 Audit & Immutable Trail

- FR-AUD-01: Every action on any PU report, user, PU master record, or configuration generates an immutable audit log.
- FR-AUD-02: Audit entries include PU Code in addition to all v1.0 fields (actor, role, timestamp, IP/device, entity, previous/new value, reason).
- FR-AUD-03: Audit logs are append-only. No user can delete or modify audit entries.
- FR-AUD-04: Correction window (default 10 minutes) allows PUAs to amend their PU report post-submission. All corrections are retained in the audit trail.
- FR-AUD-05: Super Admin can export audit logs filtered by PU Code, Ward, LGA, user, date, and action type.

6. Dashboards & Analytics

6.1 Global Filter Bar (v2.0 - PU Filter Added)

Filter	Options / Input	Scope
Date / Time Range	Date picker (from / to)	All dashboards
Submission Method	Multi-select: Web, App, SMS	All dashboards
LGA	Dropdown (27 LGAs)	All dashboards
Ward	Dropdown (287 Wards, filtered by LGA)	All dashboards
Polling Unit	Searchable dropdown (all PUs in selected Ward; ~3,459 total)	All dashboards (NEW in v2.0)
Party	Dropdown (master party list)	Reports, Candidates, Results
Senatorial District	Dropdown	Reports, Candidates
Candidate	Dropdown (filtered by party)	Candidate dashboard
Gender	Multi-select: Male, Female, Other	Agent dashboard
Login Status	Dropdown: Logged In / Not Logged In	Agent/PUA status panel
PU Submission Status	Multi-select: Submitted / Pending / Flagged / Escalated / Unassigned	PU Coverage dashboard (NEW)
Security Present	Toggle: Yes / No / All	Security incidents panel
Report Status	Multi-select: Draft / Submitted / Verified / Flagged / Escalated	All report views

6.2 Required Dashboard Panels (v2.0 Updates)

Panel 0 (NEW): PU Coverage Overview (Primary Dashboard)

PU-Level: This is the primary situational awareness panel introduced in v2.0. It provides real-time visibility of PU reporting completion across all Wards and LGAs.

- State-level coverage counter: "X of 3,459 PUs Submitted" with percentage and progress bar
- Per-LGA coverage table: LGA name, total PUs, submitted, pending, flagged, escalated
- Per-Ward coverage: click any LGA row to expand into its Wards; click any Ward to see individual PU statuses
- PU status chips per row: Submitted (green), Pending (grey), Flagged (yellow), Escalated (red), Unassigned (orange)
- Late PU alert: PUs that have not submitted by the configured threshold time are highlighted in amber

Panel 1: PUA Login Status

- Filter dropdown: Logged In / Not Logged In; count per LGA, Ward, and PU

- List of PUAs who have not logged in, with their assigned PU Code and phone number for outreach
- Last login timestamp per PUA

Panel 2: Submissions by Polling Unit (NEW granularity)

- Sortable table: PU Code, PU Name, Ward, LGA, Agent Name, Submission Time, Status, Party Results (summary)
- Click row to open individual PU report detail view
- Export this table as CSV or PDF with all filter metadata

Panel 3: Submissions by Ward (Aggregated from PUs)

- Ward-level rollup showing: Ward Name, LGA, total PUs in ward, PUs submitted, aggregate vote totals, status breakdown
- Coverage indicator: "10/12 PUs Submitted" shown per Ward row
- Click Ward row to drill down to its individual PU reports

Panel 4: PUAs by Gender

- Donut chart: Male / Female / Other counts for PUAs; filterable by LGA and Ward

Panel 5: By Local Government Area (PU Resolution)

- Choropleth map or sortable list showing LGA-level metrics with PU coverage percentage
- Click LGA to drill: LGA > Ward list > PU list > Individual PU report

Panel 6: By Political Party (PU-Level Results)

- Stacked bar chart: vote totals per party across selected PUs / Wards / LGAs
- Based on Part D party result entries from individual PU reports
- Note: only PUs that have submitted Part D results contribute to this panel

Panel 7: By Senatorial District

- Aggregated key metrics per district: total PUs, PUs submitted, % coverage, total valid votes, total invalid votes

Panel 8: Candidate Distribution (PU-Level Source)

- Per-candidate vote totals sourced from PU-level party result entries
- Sortable by votes, party, Ward, LGA, Senatorial District

Panel 9: Security Incidents (PU-Precise)

- List and map of PU reports where security_present = true
- Map pins placed at PU GPS coordinates if available, otherwise at Ward centroid
- Breakdown by security type; severity colour-coding for escalated PUs

Panel 10: Media Gallery

- Grid view of media attachments per PU report; shows PU Code label on each card
- Recommended: EC8A result sheet photos given highest visual prominence

Panel 11: Real-Time Feed / Inbox (Supervisor-Facing - PU Level)

- Live queue of incoming PU reports and SMS; each entry shows PU Code, PU Name, Ward, Agent, Timestamp, Status chip
- Quick actions: Verify, Flag, Escalate (includes PU Code in escalation record), Call Agent, SMS Reply

6.3 PU Coverage Drill-Down Navigation

All dashboard panels support a unified PU-resolution drill-down path:

- State Overview --> LGA --> Ward --> Polling Unit --> Individual PU Report
- Maximum 4 clicks from State Dashboard to viewing a specific PU report
- Breadcrumb trail maintained at every level
- Filter state preserved through the drill-down navigation

6.4 Print & Export (v2.0 - PU Dimension Added)

All eight standard report dimensions available as PDF or CSV (seven from v1.0 + one new PU dimension):

- Agents (PUAs) list by Polling Unit (NEW)
- Agents list by Ward (aggregated from PU assignments)
- Submissions by Gender
- Submissions by Polling Unit (NEW)
- Submissions by Ward
- Submissions by Local Government
- Submissions by Party (PU-sourced results)
- Submissions by Senatorial District
- Candidate Distribution across Polling Units (NEW granularity)

All exports include: Situation Room name, applied filters (including PU filter if active), generation timestamp, generator name, and signature block. PDF generated server-side with headers, footers, and page numbers.

7. Data Model

7.1 Core Entities (v2.0 - polling_units Added)

Entity	Key Fields	Description
states	id, name	State-level reference (single record: Jigawa State)
senatorial_districts	id, name, state_id	Senatorial zones for aggregated reporting
lgas	id, name, state_id, supervisor_ids[]	27 LGAs with assigned supervisor contacts
wards	id, name, lga_id, senatorial_district_id, total_pu_count	287 Electoral Wards. total_pu_count maintained for coverage calculations.
polling_units (NEW)	id, inec_pu_code (unique, format: JG/LG/WW/PPP), name, ward_id, lga_id, registered_voter_count, gps_latitude, gps_longitude, is_active, assigned_pua_id	Atomic reporting unit. Approx. 3,459 PUs in Jigawa State. One PUA assigned per PU.
parties	id, name, acronym, colour_hex	Master political party registry
candidates	id, name, party_id, post_type, ward_id	Candidates per post type and ward/LGA
users	id, full_name, phone, role_id, lga_id, ward_id, polling_unit_id (new), gender, party_affiliation, device_id, last_login, is_active	All system users. PUAs have polling_unit_id set (non-null for Layer 3).
pu_reports (replaces ward_reports)	id, pua_user_id, polling_unit_id, ward_id, lga_id, inec_pu_code (denormalised for query performance), party_agent_id, inec_presiding_officer_name, registered_voter_count (snapshot), accredited_voters, security_present, security_types[], voting_closed_time, number_cast_votes, voting_papers_unused, valid_ballot_papers, invalid_ballot_papers, ec8a_serial_number, turnout_rate (computed), remarks, status, submitted_via, device_id, sync_status, submitted_at_device, created_at	Core PU-level election report. One record per PU per election.
pu_party_results (NEW)	id, pu_report_id, party_id, votes_scored, created_at	Party-level vote results per PU. One row per party per PU report.
report_media	id, pu_report_id, media_type, storage_url, thumbnail_url, is_sensitive, upload_status, chain_of_custody_notes	Media evidence attached to PU reports

Entity	Key Fields	Description
audit_logs	id, actor_user_id, actor_role, action_type, entity_type, entity_id, polling_unit_id (new), inec_pu_code (new), previous_value, new_value, reason, ip_address, device_info, created_at	Immutable audit trail including PU context
sms_messages	id, direction, from_phone, to_phone, body, parsed_pu_report_id, delivery_status, sent_at	SMS transaction log referencing PU reports
ward_aggregates (materialised view)	ward_id, total_pus, pus_submitted, pus_pending, pus_flagged, total_accredited, total_cast, total_valid, total_invalid, computed_at	Auto-computed Ward rollup from PU reports for dashboard performance

7.2 PU Report Status Lifecycle

Status	Description	Who Can Set
draft	Saved on PUA device; not yet synced to server	System (auto)
submitted	Successfully received by server; PU coverage count incremented	System (auto on receipt)
verified	Supervisor confirmed PU report accuracy	Supervisor, Super Admin
flagged	Marked for attention at PU level; reason required	Supervisor, Super Admin
escalated	Elevated PU incident; triggers notifications with PU Code	Supervisor, Super Admin

7.3 Aggregation Logic

> **PU-Level:** Ward aggregates are computed automatically from pu_reports. The ward_aggregates materialised view is refreshed every 60 seconds during election windows and every 10 minutes outside election windows.

- total_cast (Ward) = SUM(pu_reports.number_cast_votes) WHERE ward_id = X AND status IN (submitted, verified)
- total_valid (Ward) = SUM(pu_reports.valid_ballot_papers) WHERE ward_id = X AND status IN (submitted, verified)
- coverage_pct (Ward) = (COUNT(submitted PUs) / ward.total_pu_count) * 100
- Party vote totals at Ward level = SUM(pu_party_results.votes_scored) WHERE party_id = P and pu_report.ward_id = W

8. API Specification (v2.0 Updates)

8.1 Polling Unit Management (NEW)

Method	Endpoint	Role	Description
POST	/api/v2/polling-units	Super Admin	Create a new Polling Unit record
GET	/api/v2/polling-units?ward_id=&lga_id=&status=	Super Admin, Supervisor	List PUs with filters (ward, LGA, assignment status)
GET	/api/v2/polling-units/:id	All roles (scoped)	Retrieve single PU record and its assignment
PUT	/api/v2/polling-units/:id	Super Admin	Update PU record (GPS, name, voter count)
POST	/api/v2/polling-units/:id/assign	Super Admin	Assign or reassign a PUA to this PU (body: {user_id, reason})
POST	/api/v2/polling-units/bulk-import	Super Admin	Bulk upload PU records via CSV; returns validation report before import

8.2 PU Reports (replaces /api/reports)

Method	Endpoint	Role	Description
POST	/api/v2/pu-reports	PUA / Supervisor / Admin	Submit new PU report (body includes Parts A-E; polling_unit_id validated against PUA assignment)
POST	/api/v2/pu-reports/batch	PUA (offline sync)	Batch submit offline-drafted PU reports
GET	/api/v2/pu-reports/:id	All roles (scoped)	Retrieve single PU report with party results and media
GET	/api/v2/pu-reports?pu_id=&ward_id=&lga_id=&party=&from=&to=&status=	Supervisor / Admin	Filtered PU report list
PUT	/api/v2/pu-reports/:id	PUA (window) / Supervisor / Admin	Update PU report with full audit trail
POST	/api/v2/pu-reports/:id/escalate	Supervisor / Admin	Escalate PU report; includes PU Code in

Method	Endpoint	Role	Description
			notification payload
POST	/api/v2/pu-reports/:id/verify	Supervisor / Admin	Mark PU report as verified; updates Ward aggregate

8.3 PU Coverage & Aggregation

Method	Endpoint	Description
GET	/api/v2/coverage/state	State-level PU coverage: total PUs, submitted, pending, flagged, escalated, unassigned
GET	/api/v2/coverage/lga/:lga_id	LGA-level PU coverage with per-Ward breakdown
GET	/api/v2/coverage/ward/:ward_id	Ward-level PU list with submission status per PU
GET	/api/v2/aggregates/ward/:ward_id	Aggregated vote totals for a Ward (computed from verified PU reports)
GET	/api/v2/aggregates/lga/:lga_id	Aggregated vote totals for an LGA
GET	/api/v2/results/party?pu_id=&ward_id=&lga_id=	Party-level vote totals at PU, Ward, or LGA granularity

8.4 Dashboards, SMS & Audit (Updated)

Method	Endpoint	Description
GET	/api/v2/dashboard/summary?from=&to=&lga=&ward=&pu=	Summary metrics; PU filter now supported
GET	/api/v2/dashboard/agents/login-status?pu=	PUA login status counts per LGA, Ward, and PU
GET	/api/v2/dashboard/security-incidents?pu=	Security incidents with PU Code and GPS coordinates where available
POST	/api/v2/sms/send	Send outbound SMS (recipients now addressable by PU Code)
POST	/api/v2/sms/receive	Inbound PU SMS webhook; parses JIG PU format
GET	/api/v2/audit?pu_code=&entity=&user_id=&from=&to=	Audit log retrieval; PU Code filter added

9. Non-Functional Requirements

9.1 Performance (v2.0 Scale Adjustments)

> **PU-Level:** With ~3,459 PUs vs 287 Wards, the system must handle approximately 12x more report records. Performance targets are adjusted accordingly.

ID	Requirement	Target
P01	API response time (reads)	< 500ms at 95th percentile under normal load
P02	PU report submission acknowledgement	< 1 second for synchronous confirmation
P03	PU Coverage dashboard render time	< 3 seconds for full state coverage panel (~3,459 PU rows)
P04	Concurrent PUA submissions (election day peak)	Sustain 3,459 near-simultaneous submissions within a 30-minute window
P05	Ward aggregate refresh latency	Ward aggregate updated within 60 seconds of any PU submission
P06	PDF export (PU-level reports)	< 45 seconds for PU-level exports with up to 3,500 rows
P07	Concurrent users (total system)	700 concurrent users (PUAs + Supervisors + Admins) without degradation

9.2 Availability & Reliability

- System uptime target: >= 99.5% during election operation windows (72 hours before through 48 hours after election day)
- System uptime target: >= 99.0% during non-election periods
- Ward aggregate materialised view must remain available even during peak write load (maintained on read replicas)
- Automated health monitoring with real-time alerts for API availability, database health, SMS queue backlog, and aggregate refresh lag

9.3 Security

- All data in transit encrypted using TLS 1.2 or higher
- PUA can only submit reports for their assigned PU — enforced at API layer AND database constraint level
- PU Code is validated against the users table assignment before any write operation is accepted
- Media files in private S3-compatible object store; pre-signed URLs with TTL
- Rate limiting: max 3 PU report submissions per PUA per hour (one PU per election; multiple allowed only for corrections within window)
- Device binding required for PUAs; unregistered devices require re-verification
- Penetration testing before go-live and after any major release

9.4 Scalability

- Database partitioned by lga_id on pu_reports table to support efficient per-LGA queries at scale
- Ward aggregates maintained as materialised views refreshed asynchronously to avoid blocking writes
- Read replicas serve all dashboard and coverage queries; primary handles writes only
- Containerised (Docker + Kubernetes / AWS ECS) with autoscaling triggered at 70% CPU or request queue depth > 100

10. Recommended Technical Stack

Layer	Technology	v2.0 Justification / Notes
Web Frontend	Laravel + Livewire (Blade)	Reactive real-time coverage panel; Livewire polling for PU status updates
Mobile App	Flutter (Android + iOS)	Native offline SQLite for PU report drafts; pre-filled PU Code in app state
Backend API	Laravel v2 (RESTful, versioned)	Versioned /api/v2/ endpoints; background jobs for aggregate refresh
Database	PostgreSQL (partitioned)	pu_reports table partitioned by lga_id; JSONB for security_types and party results
Aggregation	PostgreSQL Materialised Views + Redis cache	ward_aggregates view refreshed via Laravel queue worker; cached in Redis for dashboard reads
Cache / Real-Time	Redis + Laravel Echo / Pusher	PU coverage counter updates pushed to Supervisor dashboards in real time
Media Storage	S3-compatible (AWS S3 / DigitalOcean Spaces)	EC8A result sheet photos stored with PU Code prefix for organised retrieval
SMS Gateway	Africa's Talking (primary) / Twilio (fallback)	JIG PU format parsed via webhook; PU Code validated before record creation
Maps / GIS	Mapbox GL JS or Leaflet.js	PU GPS coordinates used for precise security incident map pins
PU Bulk Import	Laravel import package + CSV validation	Admin bulk-uploads INEC PU list; format and duplicate validation before commit
PDF Generation	wkhtmltopdf or DomPDF (Laravel)	PU-level report exports with INEC PU Code in all headers
Monitoring	Prometheus + Grafana	PU submission rate, aggregate refresh lag, and SMS queue added as custom metrics

11. Escalation Matrix & Notifications

11.1 Three-Tier Escalation Framework (v2.0 - PU Code in All Records)

> **PU-Level:** All escalation records now include the INEC PU Code and PU Name as mandatory fields, enabling precise geo-referencing of every incident.

Level	Actor	Timeline	Actions Required	Channels
L1	Supervisor	Within 10 min of PU flag	Call PUA; confirm PU-level facts; update annotation with PU Code; SMS to LGA Coordinator	In-app, SMS
L2	LGA Coordinator	Within 30 min	Coordinate with security actors at PU location; prepare consolidated LGA incident report; escalate to Layer 1 if unresolved	SMS, Email, Phone
L3	State / Incident Response	Immediate for violence / major incidents	SMS + Email + Phone to State Security and Steering Committee with PU Code and GPS coordinates; prepare public advisory if required	SMS, Email, Phone, Push

11.2 Automated Notification Rules (v2.0)

- PU report with security_present = true submitted: Supervisor receives immediate in-app notification with PU Code highlighted
- PU with turnout_rate > 100% submitted: system auto-flags the PU report and notifies Supervisor for immediate review
- PU report party result sum mismatch submitted and annotated as "acknowledged": Supervisor receives low-priority flag for review
- PU not submitted within configurable threshold (default: 30 min before correction window close): LGA Supervisor receives "Missing PU" alert with PUA contact
- Supervisor escalates a PU to Level 2 or 3: system auto-sends SMS and email to escalation contacts including PU Code and name
- System health breach: Super Admin and tech support auto-alerted for API latency > 2s, SMS queue > 50 pending, aggregate refresh lag > 5 min

12. Rollout Plan & Implementation Milestones

12.1 Phased Rollout Strategy (v2.0)

Phase	Scope	Key Deliverables & Acceptance Criteria
Phase 0	Pre-Development	PU master list sourced from INEC (all ~3,459 PUs with INEC codes); Role matrix; data dictionary; SMS JIG PU format guide; SOPs; UAT checklist; DB schema seeded with LGAs, Wards, and PUs
Phase 1	Core Build (API + Web)	Auth + roles; PU management module; PU report API (Parts A-E); PUA assignment; supervisor inbox at PU level; audit logging; ward aggregate computation
Phase 2	Flutter App + SMS Gateway	Offline PU draft + sync; pre-filled PU Code in app; JIG PU SMS parser + confirmation; outbound SMS by PU grouping; mobile form parity
Phase 3	Analytics + Reports	PU Coverage dashboard (Panel 0); all 11 dashboard panels; all 9 export report types; PDF print; global filters including PU filter
Phase 4 (Pilot)	2 LGAs (1 urban, 1 rural) - all PUs within selected LGAs	PUA enrolment at PU level; UAT; offline sync; SMS JIG PU parsing; PU coverage tracking; Ward aggregate validation; acceptance criteria met
Phase 5 (Waves)	Remaining 25 LGAs in 4 waves (5-7 LGAs / wave)	PU-level enrolment; training delivery per wave; wave-by-wave acceptance sign-off before next wave
Phase 6 (Go-Live)	All 287 Wards / ~3,459 PUs	All PUAs enrolled and tested; 100% PU coverage target met in enrolment; MFA enabled; 24/7 support active; escalation contacts live

12.2 Pre-Go-Live Operational Checklist (v2.0)

1. **All ~3,459 PUs loaded in system with valid INEC PU Codes, Ward, and LGA mappings**
2. **Each PU has an assigned and verified PUA (zero "Unassigned PU" count on Coverage Dashboard)**
3. All PUAs completed at least one test PU report submission (including offline sync test)
4. All Supervisors assigned to LGAs with PU coverage visibility confirmed in their account
5. SMS JIG PU format tested: end-to-end round-trip with PU Code validation from 5 diverse devices
6. Flutter app installed and offline-draft sync tested on low-end Android devices used by PUAs
7. Super Admin MFA enabled; PU master data locked for election day (no structural changes after lock)
8. Escalation contact list published with PU Code lookup procedure documented
9. Backup and restore procedure validated against staging database with full PU data
10. Monitoring dashboards active: uptime, API latency, Ward aggregate refresh lag, SMS queue, PU coverage count
11. Security penetration test completed; all critical findings remediated
12. Ward aggregate materialised view load-tested with simulated 3,459 concurrent PU submissions

13. KPIs & Success Metrics

13.1 Operational KPIs (v2.0)

#	KPI	Target	Source
1	% of PUs with an assigned and verified PUA before election day	100%	polling_units.assigned_pua_id
2	% of PUs that have submitted a report by poll close	>= 95%	pu_reports count vs total PUs
3	% of PUAs logged in at least once before election day	>= 98%	users.last_login
4	Avg. time: PU report submission to Supervisor acknowledgement	< 5 minutes	pu_report vs audit_log timestamps
5	% of PU reports with at least one media attachment (EC8A photo)	>= 75%	report_media count per pu_report
6	Ward aggregate refresh latency	< 60 seconds	ward_aggregates.computed_at lag
7	L1 escalation response time	100% within 10 min	escalation_log timestamps
8	SMS inbound PU parsing success rate (JIG PU format)	>= 90%	parsed_pu_report_id not null
9	System uptime during election window	>= 99.5%	Prometheus / Grafana
10	% of PU reports with Party D results completed	>= 80%	pu_party_results count per pu_report

13.2 Data Quality KPIs

- % of PU submissions with turnout > 100% (auto-flagged): Target < 1%
- % of PU submissions with party result sum mismatch (valid != sum of party votes): Target < 5%
- % of PU submissions failing cross-validation (valid + invalid > cast votes): Target < 3%
- Average edits per PU report post-submission: Target < 1.2
- Discrepancy rate in random audit sample of PU reports: Target < 3%

14. Risks & Mitigation

#	Risk	Likelihood	Impact	Mitigation Strategy
R1	Incomplete PU master list: INEC PU codes not available before development	Medium	High	Engage INEC liaison early; use INEC public PU register as fallback; provide bulk-import tool for late updates
R2	Insufficient PUAs: not enough trained agents to cover all ~3,459 PUs	Medium	High	Coverage dashboard flags unassigned PUs immediately; Supervisor can submit on behalf of PU; SMS fallback available for last-resort PU reporting
R3	Poor network in rural PUs prevents submission	High	High	Offline-first Flutter app with auto-sync + SMS JIG PU fallback; draft saved with INEC PU Code for accurate attribution on sync
R4	PUA data entry errors in party result totals (Part D)	Medium	Medium	Real-time party vote sum counter on form; warning + soft-block on mismatch; EC8A photo attachment as cross-reference
R5	Ward aggregate computation lag causing stale dashboards	Low	Medium	Materialised view refresh every 60 seconds; Redis cache layer for reads; monitoring alert if refresh lag > 5 minutes
R6	SMS JIG PU format non-compliance by PUAs in the field	Medium	Low	SMS training with laminated PU-specific cheat card per agent; auto-reply provides PUA's own PU Code if format is wrong
R7	Unauthorised access or data breach at PU record level	Low	Critical	API and DB enforce PU-level row security; PUA API rejects any submission for non-assigned PU; MFA for high-privilege users; pen testing before go-live
R8	System overload from ~3,459 near-simultaneous PU submissions at poll close	Medium	High	Load tested with simulated 3,459 concurrent submissions; horizontal autoscaling; async aggregate refresh; database partitioned by lga_id
R9	Intentionally fabricated PU reports	Medium	High	EC8A photo requirement; PU Code tied to specific geographic location; random audit sampling; cross-reference with INEC ward collation sheets
R10	GPS coordinates unavailable for rural PU incident mapping	High	Low	GPS optional; fall back to Ward centroid on maps; encourage PU GPS capture during enrolment when PUAs visit their PU site

15. Governance, Training & SOPs

15.1 Steering Committee

- State Chairman (Chair): final authority on policy decisions and public communications
- INEC Liaison Representative: source of official PU list; alignment on EC8A cross-referencing
- Security Coordinator: escalation matrix oversight; PU-level incident response coordination
- Civil Society Representative: independent oversight and accountability
- Technical Lead(s): system availability, PU master data integrity, aggregate computation
- Communications Lead: public release policy and official summaries

15.2 SOPs to Develop (v2.0 Additions)

13. PU Master Data Onboarding: procedure for sourcing, validating, and loading the INEC PU list
14. PUA Assignment & Deactivation: end-to-end process for assigning PUAs to specific PUs, verifying assignments, and handling reassignments on election day
15. PU Report Submission & Correction Policy: PUA workflow, party result entry, EC8A photo guidance, correction window rules
16. Escalation Ladder (PU-Precise): roles, timelines, PU Code in all communications, and decision authority
17. Media Handling: EC8A result sheet photo requirements, sensitive content tagging, chain-of-custody for PU-level evidence
18. Communications & Public Release: criteria for releasing PU-sourced aggregates publicly

15.3 Training Programme (v2.0)

Role	Duration	Topics Covered
Polling Unit Agent (PUA)	90 min	Account setup; pre-filled PU Code; form Parts A-E including party result entry; EC8A photo guidance; SMS JIG PU format; offline sync; escalation contacts
Supervisor	2.5 hours	PU Coverage dashboard; PU Inbox workflow; PU-level validation/flag/escalate; Ward aggregate interpretation; missing PU alerts; PU SMS management; report export at PU level
Super Admin	3.5 hours	PU master data management; bulk import; PUA assignment; full dashboard configuration; PU-level audit log review; aggregate refresh management; official PU-level report generation

Training deliverables:

- Laminated PU-specific cheat card per PUA: their assigned PU Code, Ward name, SMS format with their PU Code pre-filled, escalation contacts
- Step-by-step illustrated PDF guide for PU report submission including Part D party results and EC8A photo
- Short video walkthroughs (5-10 minutes per key workflow)
- In-app FAQ accessible from any screen without leaving the app

16. Appendices

Appendix A: SMS Format Reference (v2.0 - JIG PU Format)

v2.0 Format	JIG PU <PUCode> <AgentInitials> <PartyCode> <Accredited> <Valid> <Invalid> <Sec:Y/N> <SecCodes>

Example	JIG PU JG/06/01/003 MA APC 120 118 2 Y POLICE,VIG

Token	Example Value	Description
<PUCode>	JG/06/01/003	Full INEC PU Code (replaces Ward Code from v1.0)
<AgentInitials>	MA	2-3 character initials matching registered PUA account
<PartyCode>	APC	3-letter party acronym from master party list
<Accredited>	120	Integer: total accredited / counted voter papers at PU
<Valid>	118	Integer: valid ballot papers at PU
<Invalid>	2	Integer: invalid/spoilt ballot papers at PU
<Sec:Y/N>	Y	Y = security present at PU; N = no security
<SecCodes>	POLICE,VIG	Comma-separated: POLICE, NSCDC, ARMY, VIG, OTHER (omit if N)

Note: Party-level vote breakdown (Part D) cannot be captured via SMS. SMS creates a partial PU report. PUA should complete full results via app or web as soon as connectivity is available.

Appendix B: INEC PU Code Format

PU Code Structure	JG / LG_NUM / WARD_NUM / PU_NUM | e.g. JG/06/01/003 = Jigawa State / LGA 06 / Ward 01 / PU 003

Segment	Value / Range	Description
State Code	JG	Fixed prefix for Jigawa State
LGA Number	01 - 27	Two-digit INEC LGA sequential number
Ward Number	01 - 40+	Two-digit ward sequential number within the LGA
PU Number	001 - 999	Three-digit sequential PU number within the Ward

Appendix C: PU Report Field Data Dictionary

Field	Type	Required	Validation Rule	Notes
polling_unit_id	Integer	Yes	Must match PUA profile assignment. API rejects mismatch.	Core PU anchor field
inec_pu_code	String	Yes (auto)	Denormalised from polling_units table. Validated against JG/NN/NN/NNN pattern.	Stored for query perf
accredited_voters	Integer	Yes	>= 0; should be <= registered_voter_count (warning)	Replaces counted_voter_papers
security_present	Boolean	Yes	true or false	---
security_types	JSON Array	Conditional	Required if security_present = true; canonical list	JSONB in PostgreSQL
number_cast_votes	Integer	Yes	>= 0	---
valid_ballot_papers	Integer	Yes	>= 0; valid + invalid <= cast_votes	Cross-field check
invalid_ballot_papers	Integer	Yes	>= 0; valid + invalid <= cast_votes	Cross-field check
ec8a_serial_number	String	No	Optional free-text; regex hint for INEC form serial format	New in v2.0
turnout_rate	Decimal	Computed	(cast_votes / registered_voter_count) * 100. Read-only.	Stored as computed field
submitted_via	Enum	Auto	web | android | ios | sms	Set by client automatically
status	Enum	Auto	draft | submitted | verified | flagged | escalated	State machine managed

Appendix D: v1.0 to v2.0 Change Summary

Area	v1.0 (Ward Level)	v2.0 (Polling Unit Level)
Reporting atom	Ward (287 entities)	Polling Unit (approx. 3,459 entities)
Field agent role	Ward Agent (one per Ward)	Polling Unit Agent / PUA (one per PU)
Report entity	ward_reports table	pu_reports table + pu_party_results table
Party results	Not captured at Ward level	Captured at PU level in Part D (pu_party_results)
Turnout rate	Not automatically computed	Auto-computed: (cast / registered_voter_count) x 100
SMS format	JIG REPORT <WardCode> ...	JIG PU <PUCode> ...

Area	v1.0 (Ward Level)	v2.0 (Polling Unit Level)
Location precision	Ward-level (no GPS)	PU-level with optional GPS coordinates from master data
Coverage tracking	Not explicitly tracked	Real-time PU Coverage dashboard (Panel 0) showing X/Y PUs submitted per Ward
Ward aggregation	Manual / one submission	Automatic: PU reports auto-sum to Ward via materialised view
API version	/api (v1 implicit)	/api/v2/ (versioned; v1 routes maintained for backward compatibility)
EC8A cross-reference	Not included	ec8a_serial_number field in Part D + EC8A photo attachment

--- End of Document ---